

Chess Thermodynamics Engine

A Statistical-Mechanics Approach to Chess Evaluation

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This document describes the design and validation of a chess engine whose position evaluation is derived entirely from thermodynamic quantities: free energy, temperature, entropy, and chemical potential.

1. Overview

Traditional chess engines evaluate positions by summing material counts and heuristic bonuses (piece-square tables, king safety, pawn structure, etc.). This engine takes a different approach: it treats the set of legal moves as a statistical ensemble and derives an evaluation from thermodynamic state functions.

The core idea is that a position with many good options is worth more than a position with a single good move. In thermodynamics, this "option value" is captured by the entropy term $T \cdot S$ in the free energy. The engine's evaluation function computes the Helmholtz free energy $F = T \cdot \ln(Z)$, where Z is the partition function over legal moves.

2. Thermodynamic Framework

The evaluation maps chess concepts onto statistical mechanics:

- Internal energy U : X-ray mobility differential between the two sides, measured in "pawn units" ($\text{PAWN_XRAY} = 3.5$ x-ray squares per pawn).
- Quality $Q(a)$: The negated child evaluation for each move a . $Q(P, a) = -F(P_a)$ in the negamax convention.
- Temperature T : The standard deviation of the Q distribution (floored at $T_{\text{FLOOR}} = 0.1$). This makes T self-normalizing -- Q_i/T always has roughly unit variance regardless of scale.
- Partition function Z : $Z = \text{SUM}_i \exp(Q_i / T)$, summing over all legal moves.
- Free energy F : $F = T \cdot \ln(Z) = \langle Q \rangle + T \cdot S$, where S is the entropy. This is the engine's position evaluation.

The key equations:

$$\begin{aligned} T &= \max(\text{stdev}(Q_1, \dots, Q_n), T_{\text{FLOOR}}) \\ Z &= \text{SUM}_i \exp(Q_i / T) \\ F &= T \cdot \ln(Z) \\ F &= \langle Q \rangle + T \cdot S \quad (\text{free energy decomposition}) \end{aligned}$$

3. The Symmetric Evaluation Problem

The free energy $F = \langle Q \rangle + T \cdot S$ includes an entropy bonus $T \cdot S$ that rewards having many good options. But in a two-player game, the opponent's options matter too. If we only credit OUR entropy but ignore the opponent's, the evaluation is asymmetric: positions where we have many moves look artificially good, even if the opponent also has many moves.

An earlier version of the engine computed `thermoSearch(g, 1)` for the eval, which only expanded OUR moves and computed OUR $T \cdot S$. The opponent's entropy was invisible. This caused the thermodynamic engine to score only 36.7% against a material-based engine with identical search.

4. X-ray Mobility as Entropy Proxy

The correct fix is to subtract the opponent's $T \cdot S$ from each child's Q value. But computing the opponent's `thermoSearch` at each leaf requires generating all of their legal moves -- a 3-4x slowdown that makes the engine unplayably slow.

The solution uses x-ray mobility as a zero-cost proxy. X-ray mobility counts how many squares each piece could reach on an empty board (the "ideal gas" approximation). The engine already computes this for `leafEval` via `fast_raw_u_split()`, which returns both sides' absolute x-ray mobility in a single board scan. No additional computation is needed.

The proxy works as follows:

- At depth 1 of `thermoSearch`, after making each move, call `fast_raw_u_split()` to get both U_w (white's x-ray) and U_b (black's x-ray).
- Q_i is set from the side-to-move's perspective using the same split values.
- The opponent's x-ray mobility is stored: `oppXray[i]`.
- After computing T from the Q distribution, adjust each Q_i by subtracting $T \cdot \ln(\text{oppXray}[i] / \text{PAWN_XRAY})$, penalizing positions where the opponent has high mobility.
- The adjusted Q values flow into the partition function Z and free energy F .

This is equivalent to approximating the opponent's entropy as $S_{\text{opp}} \sim \ln(\text{effMoves})$ where $\text{effMoves} = \text{oppXray} / \text{PAWN_XRAY}$, and subtracting $T \cdot S_{\text{opp}}$ from the evaluation. The x-ray count divided by 3.5 (the average x-ray mobility of a pawn) converts raw x-ray squares into an effective move count.

5. Implementation

The complete thermoSearch function:

```
function thermoSearch(g, depth) {
  nodeCount++;
  if (depth <= 0) return leafEval(g);
  const moves = g.fast_moves();
  const n = moves.length;
  if (n === 0)
    return g.fast_in_checkmate() ? -CHECKMATE_SCORE : 0;

  const Qs = new Array(n);
  let oppXray;
  if (depth === 1) oppXray = new Array(n);

  for (let i = 0; i < n; i++) {
    g.fast_make(moves[i]);
    if (depth === 1 && g.fast_raw_u_split) {
      const s = g.fast_raw_u_split(XRAY);
      const t = g.turn();
      Qs[i] = t==='w' ? s.Ub-s.Uw : s.Uw-s.Ub;
      oppXray[i] = t==='w' ? s.Uw : s.Ub;
    } else {
      Qs[i] = -thermoSearch(g, depth - 1);
    }
    g.fast_undo();
  }

  // Temperature from quality distribution
  let sum = 0;
  for (let i = 0; i < n; i++) sum += Qs[i];
  const mean = sum / n;
  let vsum = 0;
  for (let i = 0; i < n; i++)
    vsum += (Qs[i] - mean) ** 2;
  const T = Math.max(Math.sqrt(vsum / n), T_FLOOR);

  // Subtract opponent entropy proxy
  if (depth === 1) {
    for (let i = 0; i < n; i++) {
      const effMoves = oppXray[i] / PAWN_XRAY;
      if (effMoves > 1)
        Qs[i] -= T * Math.log(effMoves);
    }
  }

  // Free energy via log-sum-exp
  let maxQ = Qs[0];
  for (let i = 1; i < n; i++)
    if (Qs[i] > maxQ) maxQ = Qs[i];
  let expSum = 0;
  for (let i = 0; i < n; i++)
    expSum += Math.exp((Qs[i] - maxQ) / T);
  return maxQ + T * Math.log(expSum);
}
```

The evalPos function wraps thermoSearch(g, 1) with a FEN-keyed cache. This evaluation runs at ~99 microseconds per call -- identical speed to the original asymmetric version, because the x-ray data

comes from the same board scan already performed for leafEval.

6. Chemical Potential Display

The engine displays a "chemical potential" for each captured piece, answering: how much did capturing this piece change the free energy? The chemical potential is defined as:

$$\mu = (F_{\text{before}} - F_{\text{after}}) + S_{\text{before}} * (T_{\text{after}} - T_{\text{before}})$$

This decomposes the impact of a capture into two terms: the direct free energy change and the entropy-weighted temperature shift. A piece whose capture dramatically changes the temperature of the position (e.g., removing the last active piece) will show a chemical potential that differs from its nominal material value.

7. Search Infrastructure

The thermodynamic evaluation plugs into a standard alpha-beta search framework with the following features:

- Iterative deepening with aspiration windows (width = 8 + depth)
- Transposition table with 16-bit Zobrist verification
- Killer move heuristic (2 killers per ply)
- History heuristic for move ordering
- Null-move pruning (R = 2)
- Late move reductions (LMR) after 4 full-width moves at depth ≥ 3
- Futility pruning (margin = 12 x-ray units)
- Quiescence search (captures, promotions, checks) to depth 3
- Delta pruning in quiescence with piece-specific deltas

All search parameters are calibrated for the x-ray mobility scale (~1 unit per pawn equivalent, ~3.5 raw x-ray squares per pawn). Node budgets range from 10K (easy) to 1M (max).

8. Validation: Fair Playtest

To validate the thermodynamic evaluation, a 30-game match was run against a material-based evaluation using identical search infrastructure. The only difference between the two engines was the leaf evaluation function.

Test conditions:

- Same search: TT, killers, history, null-move, LMR, futility, quiescence, aspiration windows
- Same budget: 30,000 nodes per move, iterative deepening
- 15 openings x 2 colors = 30 games (each opening played from both sides)
- Material eval: standard piece values (P=1.0, N=3.2, B=3.3, R=5.0, Q=9.0)

Results:

Engine	Wins	Draws	Losses	Score
Thermo (symmetric)	6	20	4	53.3%
Material	4	20	6	46.7%

By color (thermo perspective):

- As White: 3 wins, 9 draws, 3 losses
- As Black: 3 wins, 11 draws, 1 loss

Comparison with the old asymmetric evaluation:

Version	W	D	L	Score
Old (asymmetric TS)	3	16	11	36.7%
New (symmetric proxy)	6	20	4	53.3%

The symmetric x-ray proxy fix improved the thermodynamic engine from clearly losing (36.7%) to slightly winning (53.3%) against material eval with identical search, a swing of +16.6 percentage points. The engine performs symmetrically by color, with no significant white/black bias.

9. Key Design Decisions

- X-ray mobility as internal energy: Instead of piece counting, the engine uses "ideal gas" piece reach (squares reachable on an empty board). This is extremely fast to compute and captures piece activity without expensive board scanning. $\text{PAWN_XRAY} = 3.5$ normalizes to pawn-equivalent units.
- Temperature from quality variance: $T = \text{stdev}(Q)$ makes the Boltzmann factor Q/T self-normalizing. Positions with widely varying move qualities have high temperature (more exploratory); positions where all moves are similar have low temperature (more deterministic).
- Free energy as evaluation: $F = T \ln(Z)$ naturally combines material advantage ($\langle Q \rangle$ term) with positional flexibility ($T \cdot S$ entropy term). A position with one great move and many bad ones evaluates lower than a position with several good options, capturing a real strategic concept.
- Zero-cost symmetric proxy: Rather than recursing to compute opponent entropy (too slow), the x-ray split from the same board scan approximates opponent move count at zero additional cost. This is the key practical innovation enabling competitive play speed.

10. Conclusion

The chess thermodynamics engine demonstrates that position evaluation can be derived from first principles of statistical mechanics rather than hand-tuned heuristics. The free energy $F = \langle Q \rangle + T \cdot S$ naturally captures both material advantage and positional flexibility in a single, theoretically motivated framework.

The symmetric x-ray proxy makes this practical: by using the same board scan to estimate both sides' entropies, the engine achieves competitive strength against traditional material evaluation with no speed penalty. The thermodynamic quantities (temperature, entropy, free energy, chemical potential) displayed during play provide genuine strategic insight rather than opaque numerical scores.